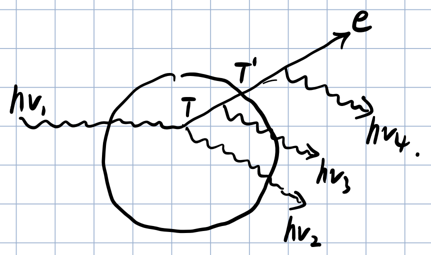
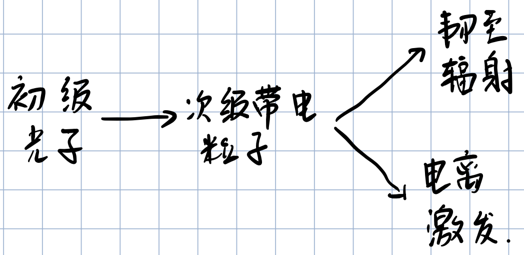




授与能:

$$\bar{\varepsilon} = (R_{in})_u - (R_{out})_u + (R_{in})_c - (R_{out})_c + \sum Q.$$

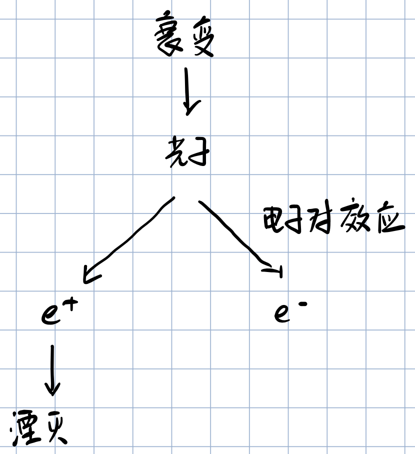
$$\bar{\varepsilon} = h\nu_1 - h\nu_2 - h\nu_3 . - T'.$$

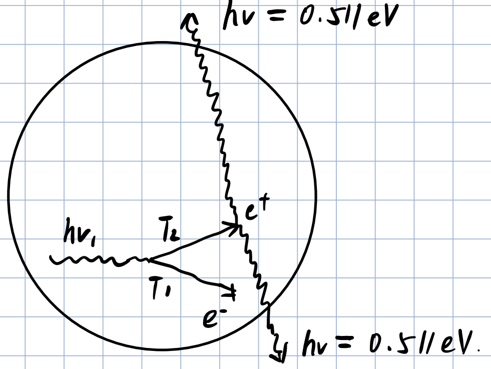
转移能

$$\varepsilon_{tr} = h\nu_1 - h\nu_2 = T$$

被转移净能量:

$$\varepsilon_{tr}^{n} = h\nu_1 - h\nu_2 - h\nu_3 - h\nu_4 .$$





$$\varepsilon_{tr} = -2h\nu + \sum Q = -1.022\ eV. + \sum Q$$

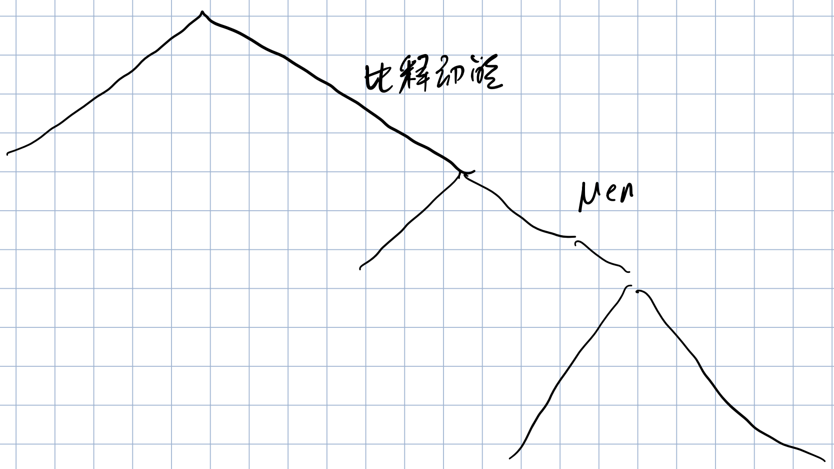
$$\varepsilon_{tr}^{n} = -2h\nu + \sum Q$$

$$\varepsilon = -2h\nu + \sum Q.$$

并且，

$$\sum Q = \underbrace{h\nu_1}_{衰变} - \underbrace{2m_e c^2}_{电子对效应} + \underbrace{2m_e c^2}_{电子湮灭} = h\nu_1.$$

$E = \frac{\rho t}{\cos\theta}$ $\frac{I\,dN}{dV}$



$\frac{dE_{tr}}{dm}$

$\frac{dE_{en}}{dm}$

$\underline{d_e}$

$\frac{d N \mu_{tr} \rho h\nu L_{\rho}}{d \Phi \rho\nu_{\nu}} = \frac{dE}{dm}$

$K_c = \frac{\mu_{en}}{\rho} \Phi h\nu$

$= \Psi \frac{\mu_{en}}{\rho}$

$$\frac{\int_E \phi' \left(\frac{S}{\rho}\right) dE}{\int_E \phi(E)\, dE}$$

$X = \frac{dQ}{dm}$

d

K_c

$\frac{e}{\bar{W}} K_c = X$

$K_c = \frac{\mu_{en}}{\rho} \Psi$

$X = \frac{\mu_{en}}{\rho} \Psi \left(\frac{e}{\bar{W}}\right)_{air}$

$\underline{K = \left(\frac{\mu_{tr}}{\rho}\right)_{E,Z.} \Psi}$

$K = \overset{?}{} \frac{X}{R}$

$K_c = \left(\frac{\mu_{en}}{\rho}\right)_{E,Z} \Psi.$

$X = K_c \left(\frac{e}{\bar{W}}\right)_{air}$

$K = \left(\frac{\mu_{tr}}{\rho}\right)_{E,Z.} \Psi.$

$= K_c \frac{\mu_{tr}}{\mu_{en}} = K_c (1-g)$

$= (1-g)\, X / \left(\frac{e}{\bar{W}}\right)_{air}$

$=$

比释动能	碰撞比释	吸收剂量	照射量
$K \equiv \frac{d\varepsilon_{tr}}{dm}$	$K_c \equiv \frac{d\varepsilon_{en}}{dm}$	$D \equiv \frac{d\varepsilon}{dm}$	$X \equiv \frac{dQ}{dm}$
$1\,Gy = 1J/kg = 100\,rad$	同 K	同 K	$1R = 2.58\,e^{-4}\,C/kg$
$K = (\frac{\mu_{tr}}{\rho})_{E,Z}\Phi$	$K_c = (\frac{\mu_{en}}{\rho})_{E,Z}\Phi$	$(\frac{S}{\rho})_{col}\Phi$	$X = \Phi \cdot (\frac{\mu_{en}}{\rho})_{air}(\frac{e}{\bar{W}})_{air}$
μ_{tr}	μ_{en}	S_{col}	$(\mu_{en})_{air}(\frac{e}{\bar{W}})_{air}$
$\forall$	$\forall$	$\forall$	Air